PROJMAN · FOUNDATIONAL SPECIFICATION

The 18-Stage Construction Lifecycle, Fully Decomposed

A complete stage-by-stage breakdown of Australian residential construction — every sub-step, responsible role, verifier, surface (App / Portal), and cross-cutting platform rule — compiled as the foundation for the App, Portal, Dashboard, Server and RDBMS specifications.

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Part 1 — Cross-Cutting Architecture (Standing Rules)

These rules were established once, across the course of the stage-by-stage design pass, and apply throughout Part 2 rather than being re-derived per stage. Where a stage table references a rule below, it is applying this section, not introducing something new.

1.1 Role model & appointment chain

Appointment chain: PM appoints Builder (Stage 9), Site Supervisor (PM's own independent on-site representative — per the locked role table, “Site Supervisor” is explicitly a subset of PM's own role, not the Builder's), and Inspector (external, per hold point). Builder appoints Foreperson and Tradies — his own crew, in his own operational scope.

Self-Registration → Introduction → Job Award (corrected, three-step, replaces the earlier two-step Register/Assign model): the original Register/Assign pattern conflated *establishing who someone is* with *committing to a specific job*. Corrected into three distinct events:

1. **Self-Registration** — every role, not only Tradie, downloads the app and builds a profile independently, at their own pace, as a platform user — unconnected to any peer, any Builder, any project. This is identity and credential-building, nothing more.
2. **Introduction** — a QR exchange, functionally a digital business-card swap between two already-self-registered users. Adds each to the other's contact/address book. No job is implied or created — this can happen years before either party has a specific project to offer the other. **Resolved:** relies on each party's own ordinary account login already established at Self-Registration, not a separate out-of-band PIN — lower friction, consistent with how a “connect” action works elsewhere. Revisit only if proximity fraud at this specific step turns out to be a real problem once real usage exists; not worth paying the friction cost upfront for a currently theoretical threat.
3. **Job Award** — a distinct, later, formal event, only at the relevant stage (e.g. Stage 9 for Builder). PM sources from people already in his own contact book — never a cold stranger — sends a real invitation, and the other party reads and signs to accept or decline. Document access for evaluation (e.g. Builder reviewing drawings/BOQ to prepare a tender quote) precedes this signature, per Stage 9's own sequencing (S9.2/S9.3 issue tender documents before S9.5's selection decision) — this was already correct and needed no change.

The same three-step shape applies one tier down, Builder↔Tradie/Foreperson, and is the standing platform primitive for every relationship in this system, not a one-off mechanic for PM↔Builder.

PM's access to Builder's output is read-only, without exception, from the moment Builder is engaged onward — his own job breakdown, his own Cost Plan, his own Programme/Line-of-Balance schedule. PM reads Builder's authored plan and progress; he never has a parallel write or override path into it. This is not a redaction of a shared record — it is a genuinely separate table/view that PM can only read, the same wall that already protects Builder's margins (Section 1.9), now stated explicitly for schedule and job-breakdown data too, since it is easy to mis-model this as “PM can read, and override if needed” — it must not carry any override path at all.

Client is portal-only, per the locked 6-role model — never paired to the App, never a sync writer.

Same-tenant model for v1: Builder is modelled as a same-tenant role for the founder's own first project (he is Builder's boss, not an arms-length counterparty). The cross-org engagement model (QR trust exchange across tenant boundaries, tenant isolation, an attested data slice flowing between independent

businesses) is real and designed in outline, but deliberately deferred until an arms-length relationship actually appears — e.g. a second Developer bringing an independently-owned Builder onto the platform.

1.2 Surface classification — App | Portal | Both

App: capture-first, offline-first (local cache, works with no signal and gloves on). Primary surface for Builder, Site Supervisor, Foreperson, Tradie, Inspector. Also used by PM for a distinct, additive purpose: a read-only recall view plus notes/voice-memo capture tied to whatever project or task is open, for use on the road — this is deliberately not full Portal functionality on a phone.

Portal: desk-analytical, connectivity assumed (though tablet-capable at the same screen quality — the dividing line is connectivity assumption and cognitive-load shape, not physical screen size). Primary surface for PM: Gantt editing, Cost Plan, tender comparison, DBYD/zoning research, client communications.

Both: simple data entry that could happen at a desk or on site — e.g. initial client/site capture at Stage 1.

Casual PM notes/voice memos captured via the App carry a timestamp and a project/task tag for recall — they deliberately do *not* receive the frozen/hashed evidentiary treatment (Section 1.4) reserved for sign-offs and prices, or every casual memo would cost as much to build and store as a legal record it was never meant to be.

1.3 Tick-then-verify (separation of duties)

Applies wherever a second, independent party exists to check the first party's work — the trigger is *independent second party*, not *site vs. office* (a distinction corrected once office-phase Stage 8 introduced an independent Inspector). A device-writable status field (the tick) is paired with a protected `verified_by` / `verified_at` field (the verify), settable only by the role one level up — identical discipline to `is_validated`.

Chain: Tradie ticks → Foreperson verifies (crew-level). Foreperson's + Tradies' output is separately overseen by Site Supervisor, who reports to PM — giving PM an independent check on Builder's crew that doesn't run through Builder himself.

1.4 Frozen / hashed record pattern

Applied to: client estimates (Stage 5), DA sign-off (Stage 7), material selections (Stage 6), progress claims, variations, final handover sign-off (Stage 18), and the PM↔Builder handover event itself (Stage 9).

Components: a content hash taken at the exact moment of send/capture; `frozen_at` timestamp; correlation to an external, independently-checkable record (e.g. the email provider's own message log) rather than resting on an internal timestamp alone; enforced immutability once frozen (a database-level trigger rejecting any UPDATE, backed by a friendlier application-level check) — corrections create a new version via `supersedes_id`, never an edit of the original.

The same discipline extends to evidence capture generally: captured inside the app only (never a gallery upload, or the timestamp/location become meaningless), signed and hashed at the moment of capture,

geofenced, append-only. This is what makes captured evidence *testimony* rather than merely a claim with a photo attached to it.

1.5 Hold point as checklist, not boolean

`hold_point_requirements` is a list per stage, not a single flag — Stage 12 alone needs three named, partly-conditional documents. Each requirement carries a `required_inspector_type` (engineer vs. building surveyor vs. certifier are not interchangeable — Stage 13 needs both an engineer and a surveyor, neither substituting for the other) and an `inspection_type` flag (`statutory` , which blocks progress, vs. `internal_qa` , which is recorded but does not block — Stages 14, 16 and 17 are internal QA only, no statutory hold point).

Two tiers, not one: a *project-level* hold point (e.g. trunk services) blocks every modular unit on the site at once; a *unit-level* hold point (e.g. one townhouse's slab) blocks only that unit. This distinction is not a schema nicety — it is what makes real Line-of-Balance, staggered multi-unit construction possible at all (Section 1.8).

1.6 Event-chaining rule

Hold point cleared → downstream stage auto-scheduled → relevant trades auto-notified. First appears explicitly at Stage 13→14 (frame validated auto-notifies electrical/plumbing that rough-in is due) and is applied as a standing pattern wherever else the same shape occurs.

1.7 Digital identity, Verified Work History & the Job Monitoring loop

Every role, not only Tradie, accrues a role-appropriate credibility record using the same primitive: Builder's is completed-project count, on-time delivery, dispute record; Site Supervisor's is inspection thoroughness; PM's is delivery reliability — making trust symmetric (Builder can check PM's reliability the same way PM checks Builder's).

Self-registration and live verification: a Tradie (or any role) builds their own profile in the App at their own pace, independent of any one Builder — licence, qualifications, and accumulated work evidence, owned by the individual. A QR scan carries only a signed reference, never the underlying data itself — the check is a live server lookup at the moment of scanning, not a photograph of a possibly-stale claim. This profile — licence status, qualifications, and accumulated work evidence — is exactly the raw material **VeriTrade** (the B2B trade-verification platform this feeds, specified separately) anchors and displays, subject to that individual's own opt-in (Section 4.2).

Evidentiary weight: voice or photo capture lowers the barrier to entry, particularly for a less digitally-comfortable worker, but a captured narrative only carries the same evidentiary weight as other testimony in this system if it anchors to an independently checkable project record (ideally corroborated by a supervising party) — otherwise it is a well-produced version of “he says,” not stronger than an ordinary review.

Anti-fraud economics: the goal is to raise the cost of forgery above its benefit, not to claim an unforgeable system. A correlated, multi-party, time-extended evidence chain is far more expensive to fabricate at scale than an isolated review (the attack surface exploited by click-farm-style review fraud). The residual risk is a slower, deliberate seeding attack — a small, real, cheaply-run job with a colluding verifier, used to mint a thin but technically genuine badge. Mitigation: weight trust by the scope and duration of verified work, not merely the presence of a badge.

Proximity/device-sharing fraud (raised in external review): a single geofenced timestamp is gameable by one worker carrying a colleague's phone to fake presence. The mitigation already exists in this design and does not need a new mechanism — it needs to be relied upon deliberately: a geofence ping is never sufficient evidence on its own, because Section 1.3's tick-then-verify chain already requires an independent Foreperson to separately verify the work was actually done. Faking proximity still has to also fool a human who is meant to be looking at the physical result, not just the phone's location — the geofence narrows the fraud window, the human verification closes it. Worth stating this explicitly rather than leaving it implicit, since it is easy to design a UI that treats a geofence match as sufficient by itself.

Fallback: manual photo/upload stays a permanent, first-class path for anyone who does not use self-service registration — not a degraded shadow of it. Forcing self-service-only onto a workforce with uneven digital access would widen exactly the inequality this design is meant to close.

Distribution: free access for TAFEs and schools seeds the network's supply side with apprentices building verified history from day one. Minors are routinely part of this population (school-based apprenticeships can start at 15–16) — this needs its own dedicated privacy/consent design, not the adult flow with an age gate attached (see Part 3).

Job monitoring loop (Portal for planning, System for comparison, App for notification): job breakdown and scheduling are authored on the Portal; monitoring is not a separate action but a live comparison of planned (Portal-authored) against actual (App-captured, via tick-and-verify); a behind-schedule or over-budget flag is the same S-curve variance logic used since the Cost Plan design, generalised to schedule; notification routing respects the same commercial visibility walls as everything else (Section 1.9) — a cost overspend flag never reaches the client, a milestone-reached flag reaches the client but without cost detail behind it.

1.8 Multi-unit / modular development model

`modular_units` (`project_id`, `unit_number`) sits between Project and Stage. Stages 1–9 are always project-level, run once regardless of whether the project is one house or sixteen. Stages 10 onward default to unit-scope, but `scope` (`project` | `unit`) lives on the individual **task**, not the stage — a single stage can mix both (trunk services are project-scope, each unit's own rough-in is unit-scope, within the same Stage 11). The Builder sets this during job breakdown at Stage 9; the system proposes sensible defaults, Builder confirms or overrides each one.

A standalone house is not a special case — it is `modular_units` with exactly one row. One model, one rule set, N ≥ 1, never a fork between “development” and “single house.”

Client modelling operates at two tiers: development-level (the whole project's principal, if any — unchanged from Stages 1–9) and unit-level (an individual buyer attached to one townhouse, who may attach before or after that unit's Stage 6 runs, or never, in a spec-built unit). Stage 6's material-selection sub-steps fire only if a buyer is attached to that unit at the time it reaches Stage 6. Billing milestones and the construction contract itself scope to the individual sold unit, not the whole development; an unsold unit is funded by developer capital or a construction finance facility instead, which needs its own tracked funding-source attribute per unit.

Execution follows Line-of-Balance / repetitive scheduling — the established construction-management technique for multi-unit work, distinct from ordinary critical-path scheduling. Crews move through units in staggered, overlapping batches (a handful of units at a time), never waiting for every unit to reach the same stage before any of them advance, and never finishing one unit completely before starting the next. This is precisely why unit-level hold points (Section 1.5) are a requirement, not a nicety — a project-only hold-point model would force exactly the lockstep waiting that real practice does not use.

Which Gantt view renders is a conditional, not a roadmap-priority decision. S1.3 has the PM declare building type/unit count at project creation, before anything else is built — that single value is what selects the tool, not an engineering guess about which project type to build for first. One unit renders the standard bar-chart Gantt; more than one renders a Line-of-Balance flow view instead. This resolves what Part 3 previously carried as an open, deferred item: the LOB rendering component still needs to be built, but the trigger for when it's needed is now exact, not a matter of when engineering gets around to it — it's required the first time any project is created with more than one unit, whenever that is, not on a fixed schedule.

1.9 Commercial visibility walls

Client sees what they are buying and what they owe — never who is paying whom upstream. The PM↔Builder↔Subcontractor commercial relationship stays invisible to the client at every tier, the same wall regardless of which tier is asking.

Governing principle for Builder ↔ PM visibility: disclosure tracks who bears the price risk, not role or employment status. Whoever absorbs the risk of a cost overrun earns the right to keep their cost structure private. Whoever is protected from that risk owes transparency in exchange — the same reasoning already applied to material-selection pricing (Stage 6), now generalised to the whole Builder relationship. Three scenarios, not one fixed rule:

1. **Independent Builder, fixed price** — Builder bears the risk (eats a blowout, keeps the upside on an efficient job), so his margins and rates stay private. PM still sees Builder's head-contract figure and his subcontractor engagement register (who is engaged, what is committed, what is owed) — not a concession, but a standing requirement regardless of pricing model, to support step-in rights if Builder becomes insolvent. Builder runs his own, separate, nested Cost Plan underneath the single head-contract line PM sees.
2. **Independent Builder, cost-plus + commission** — once PM/client is absorbing actual cost rather than Builder bearing price risk, hiding the cost structure would let Builder extract margin with no accountability for it — the same conflict-of-interest shape flagged earlier for undisclosed referral commissions. Full disclosure of cost *and* commission is what makes cost-plus an honest pricing model rather than a fixed price wearing a disguise. This extends to job costing and the supplier/subcontractor register in full, not just the head-contract figure.
3. **Employee Builder** — not a separate entity at all, so there is no commercial wall to begin with: there is no second party's margin to protect. Every dollar spent on subcontractors is the company's own money in the company's own books. This is not "Builder manages his own finances within a budget" — there is only **one** Cost Plan, not a private one nested under a head-contract line; an employee Builder has spending authority within it, not a separate ledger. Full visibility here isn't extracted from him, it's simply what "not a separate entity" already means.

This does not conflict with Section 1.1's rule that an employee Builder doesn't get full portfolio financial visibility — that rule is about Builder looking *outward/upward* into the company's other projects, which he still shouldn't see. This rule is about the company looking *inward* at a project it is running through its own employee, which it obviously can. Different direction, same architecture, not a contradiction.

Subcontractor pass-through consent, required in scenarios 2 and 3: a subcontractor agrees to disclose pricing to Builder, not necessarily to PM — a party he has no direct contract with. Builder's own subcontractor agreements need an explicit clause permitting this upward disclosure wherever scenario 2 or 3 applies, mirroring whatever Builder himself agreed to with PM. Without it, the transparency built for the Builder–PM relationship quietly becomes an unconsented privacy problem for a third party who was never part of that agreement.

Concretely: a `builder_engagement_type` field (`employee` | `independent_fixed` | `independent_cost_plus`) on the engagement record drives which of the three visibility scenarios above applies — a per-engagement setting, not a role check, since the same person could be an employee on one project and an independent cost-plus Builder on another.

The Builder–PM pricing model is independent of the client-facing pricing model — PM can run cost-plus with Builder while still quoting the client a fixed price on the house, or any other combination. Choosing one does not move the other; they are two separate walls, decided separately.

Cost-plus vs. fixed pricing is a client choice per selection line item, not a whole-of-contract setting — only the cost-plus option ever discloses `actual_cost`.

DA-phase consultant costs (Architect, Town Planner, Engineers) are absorbed by the company rather than billed to the client, made financially sound specifically because standard product tiers use a pre-negotiated `rate_card` with the Architect rather than a bespoke fee negotiated per project — the company knows its own product well enough to bound the cost. Tracked internally per project regardless (`billable_to_client = false`) to measure true cost-to-win and conversion rate across the pipeline.

1.10 Billing milestones

Milestone	Stage	Trigger
Deposit	9	Construction contract signed / Builder engaged
Base	12	Slab validated (all hold points cleared)
Frame	13	Frame validated (engineer + surveyor + timber cert)
Lock-up	14	Building confirmed secured from weather
Fixing	16	Final trade sign-off, per room
Completion	18	Project status set to completed

Only these six stages carry a client-facing billing event, using the frozen/ hashed pattern (Section 1.4). The client sees no cost or schedule internals between milestones — a coarse status view only, consistent with Section 1.9.

1.11 Handover documentation (as-built record)

The end-of-project handover pack is a **query**, not an authoring task — every `inspections`, `certificates`, `deliveries` and `material_selections` row already tied to the project/unit's stage-and-task tree, since Stage 10, composes it directly. This is only true because of the evidentiary discipline enforced at every hold point along the way (Section 1.4/1.5); a system that only logged status without evidence could not make this claim.

This data needs its own long, probably indefinite, retention class — distinct from, and not swept up by, any routine log-pruning policy designed for security/access logs. Structural defect liability periods commonly run years in Australia; the exact figure varies by state and defect category and should be confirmed against current legislation, not assumed.

Scoped per `modular_unit` — each townhouse in a development gets its own, correctly scoped handover pack, since each is sold and warrantied independently.

Part 2 — The 18 Stages, Fully Decomposed

Each stage lists its sub-steps with Responsible party, Verifier (where tick-then-verify applies), Surface, and Notes. A stage-level note above the table records any correction made during design or a cross-cutting rule the stage first established — see Part 1 for the full rule.

A — PRE-DESIGN & FEASIBILITY

STAGE 1 PROJECT CREATION LAND INGESTION

> Office stage, single actor (PM). Gated on completeness of required fields and documents, not tick-then-verify.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S1.1	Capture client profile — name, contact, billing address, ABN (if company)	PM	—	Both	writes `customers`
S1.2	Capture site details — address, lot/plan number, title reference	PM	—	Both	writes `projects`
S1.3	Declare building type / unit count — single dwelling, or multi-unit development (and how many units)	PM	—	Both	writes `modular_units` (Section 1.8) — this single declaration is what conditionally selects every downstream tool: standard bar-chart Gantt for a single dwelling, Line-of-Balance flow view for a multi-unit development (Section 1.8/Part 3). Not a roadmap-priority question — the Portal renders whichever the project actually needs, driven by this value, not by which one engineering happened to build first.
S1.4	Upload land title document	PM	—	Both	`documents.kind='land_title'` ; app camera capture is primary use case
S1.5	OCR extract — owner name(s), lot boundary (m²), perimeter length	System	—	Both	fires wherever S1.4 happened
S1.6	Upload contour plan / land survey report	PM	—	Both	`documents.kind='survey'`
S1.7	Confirm / override extracted OCR values	PM	—	Both	PM may check on-site or back at office
S1.8	Generate tracking ID + confirmation email to client	System	—	—	itemised file list, timestamped receipt

A — PRE-DESIGN & FEASIBILITY

STAGE 2 ENVIRONMENT UTILITY HAZARD AUDIT

> Office stage, desk research. Establishes the standing rule: any AI-generated output reaching a client requires PM confirmation first (S2.6) — applied at every subsequent AI-generation stage without re-asking.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S2.1	Submit Dial-Before-You-Dig referral; record utility coordinates, sewer depth, overhead power clearance	PM	—	Portal	requires document — DBYD response
S2.2	Record council zoning code; cross-check planned scope against density allowance	PM	—	Portal	
S2.3	Record easements, restrictive covenants, council design guidelines	PM	—	Portal	requires document if separate from land title
S2.4	Record environmental hazard buffers — flood zone, heritage overlay, bushfire-prone area	PM	—	Portal	**Resolved:** this finding is the flag source for downstream conditional requirements (Stage 12's compaction/compression tests, Stage 16's energisation certificate) — a hazard recorded here auto-triggers the relevant later requirement rather than each stage re-asking independently.
S2.5	AI/NLP compliance check against local planning laws + building codes	System	—	—	
S2.6	PM reviews and confirms/overrides the AI audit result before it goes anywhere	PM	—	Portal	**standing rule origin** — AI output never reaches a client unconfirmed
S2.7	Auto-send audit summary email to client	System	—	—	fires only after S2.6

A — PRE-DESIGN & FEASIBILITY

STAGE 3 CONCEPT DESIGN GENERATION

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S3.1	PM triggers concept generation (one click)	PM	—	Portal	pulls zoning + boundary data from Stages 1–2
S3.2	AI computes structural footprint within setback-compliant envelope	System	—	—	

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S3.3	AI overlays mandatory green open-space zones	System	—	—	
S3.4	AI generates elevation profile (max height)	System	—	—	
S3.5	AI calculates internal floor area summary (m²)	System	—	—	
S3.6	AI generates setback-rationale text	System	—	—	
S3.7	PM reviews and confirms/overrides before anything reaches the client	PM	—	Portal	standing confirm-gate rule
S3.8	Store 3 PDFs (Footprint, Elevation, Area Summary) + rationale to documents repository	System	—	—	`documents.kind='concept_design'`
S3.9	Auto-email client with attachments	System	—	—	**Resolved** — see S3.10/S3.11 for the revision loop
S3.10	PM records requested changes from client (call/meeting/email) and re-triggers generation	PM	—	Portal	no client portal write here — pre-contract, PM still the sole recorder per the Stage 4 correction
S3.11	AI regenerates concept design as a new version	System	—	—	new version linked to the original via `supersedes_id`, same versioning discipline as every other frozen record (Section 1.4) — loops back through S3.7's confirm-gate and S3.8's storage before re-emailing; original version never overwritten

A — PRE-DESIGN & FEASIBILITY

STAGE 4 TOWN PLANNER COUNCIL SCREENING

> Corrected during design: the matrix originally described a client acceptance step here. Removed — this stage is pre-contract; the project-developer-runs-DA-alone business model means no client sign-off or billing belongs at this point.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S4.1	PM selects screening pathway — engage an independent Town Planner, or submit a Preliminary Construction Intent direct to council	PM	—	Portal	genuine branch decision, not automatic
S4.2	System packages Stage 3 data assets for transmission	System	—	—	
S4.3	System drafts and sends the email to whichever recipient was chosen at S4.1	System	—	—	
S4.4	Calendar countdown starts, tracking expected Town Planner/ council response time	System	—	—	**Resolved:** same mechanism as Stage 7's statutory clock, applied here too — not a formal statutory deadline in every case, but PM benefits from the same visibility regardless of whether the countdown is a legal obligation or just an expected turnaround
S4.5	PM records the outcome received	PM	—	Portal	no billing trigger — pre-contract

A — PRE-DESIGN & FEASIBILITY

STAGE 5 BUDGET BASED STYLE GENERATION

> Introduces the frozen/hashed record pattern, reused at every later point where a client-facing figure or document must be provably unforgeable: a content hash taken at the moment of send, the row locked (no UPDATE once frozen — enforced by a DB trigger, not just convention), correction via a new superseding version rather than an edit, and the send event correlated to an external record (e.g. the email provider's own message log) rather than resting on an internal timestamp alone.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S5.1	Style selector unlocks	System	—	—	gated on Stage 4 outcome
S5.2	AI generates 3 cost-calibrated options (Budget / Deluxe / Premium)	System	—	—	
S5.3	PM reviews and confirms/ overrides before it goes out	PM	—	Portal	standing confirm-gate rule
S5.4	System emails client the 3 options with specs + cost models	System	—	—	`estimate_lines` frozen at send: `content_hash`, `frozen_at`, `email_message_id`, `supersedes_id` for corrections. Pre-contract, informational — no client portal write yet.

STAGE 6 ARCHITECTURAL DRAWING DEVELOPMENT

First stage with a real, narrow client-portal write — distinct from, and not in conflict with, the rule that the client doesn't need construction-progress or job-costing visibility. Picking a tier and choosing materials is a purchase decision, not a build-progress one. First stage confirming the actor-allocation convention: external professionals (draftsman/architect) never get a login — PM is the sole system user through Stage 7, receiving their work externally and uploading it.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S6.1	Capture client's selected budget tier (from Stage 5's 3 options)	Client	—	Portal	narrow client write
S6.2	Route selected tier's data files to the appointed draftsman/ architect	PM	—	Portal	external party, no login
S6.3	Client uses interactive material selection (tapware, tiles, paint, fixtures)	Client	—	Portal	flagged as its own module deserving dedicated design (see Part 3) — not fully specified here. Each line carries `pricing_model` (fixed | cost_plus, client's choice), `allowance_amount`, `actual_cost` (cost-plus only), `variance_band_pct`, `price_lock_expiry`. Two freeze points per line: `selected_at` (intent) and `committed_at` (re-verified against supplier immediately before the item is actually ordered/paid — mirrors the job-costing rule that 'committed' means PO issued, not preference expressed). Stock unavailable or price outside the variance band at commit time → routed through the existing Variation mechanism, not a new one.
S6.4	*(external)* Draftsman/ architect completes DA drawing set	—	—	—	no system actor, external turnaround time only
S6.5	PM uploads finalized structural floor plans + elevation views	PM	—	Portal	`documents.kind='architectural_drawing'`

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S6.6	System auto-maps client's material selections onto uploaded drawings → 3D interior preview	System	—	—	depends on both S6.3 and S6.5
S6.7	PM reviews 3D preview before it reaches the client	PM	—	Portal	standing confirm-gate rule
S6.8	Client views 3D preview	Client	—	Portal	view only, not a write

A — PRE-DESIGN & FEASIBILITY

STAGE 7 DA SUBMISSION APPROVAL

> First stage where a client action carries real legal weight rather than a preference — the frozen/hashed discipline from Stage 5 applies with more force here, since this is the client legally approving a professional deliverable before it goes to council.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S7.1	PM does final release check before the DA package goes to the client	PM	—	Portal	standing outbound-release discipline
S7.2	System transmits the complete, certified DA drawing set to the client for review	System	—	—	fires only after S7.1
S7.3	Client reviews and provides digital sign-off	Client	—	Portal	`content_hash` against the exact DA package version; `frozen_at` the instant the signature lands; no update path, ever. **Resolved:** if the client declines, it routes back to S6.2 for redraft, billed as a Variation against the architect's redraft cost — no new mechanism, reuses the existing Variation workflow.
S7.4	System compiles the final certified drawing package	System	—	—	only after S7.3 approval
S7.5	System submits to council / certifying authority	System	—	—	status → `DA_PENDING_COUNCIL_REVIEW`

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S7.6	Calendar countdown starts, tracking council's statutory response clock	System	—	—	
S7.7	Auto-email confirms DA submission	System	—	—	to PM at minimum

B — APPROVALS & CONTRACT

STAGE 8 DA APPROVAL CONSTRUCTION CERTIFICATE

> First stage with a genuine second, independent actor — corrects the Stage-1 rule that office stages don't need tick-then-verify separation of duties. The real trigger was always 'single actor vs. independent second party', not 'office vs. site'; this is the first office-phase stage to break that pattern.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S8.1	Record DA approval received from council; status → `DA_APPROVED`	PM	—	Portal	upload approval document
S8.2	Engage Principal Certifier (independent building surveyor)	PM	—	Portal	`consultants.discipline='certifier'`
S8.3	Structural engineer certifies plans under AS3600/ AS3850	Structural Engineer (external, no login)	—	—	PM uploads the certificate on receipt
S8.4	Prepare and lodge Construction Certificate application	PM	—	Portal	
S8.5	**HOLD POINT** — Principal Certifier verifies NCC compliance	Inspector	protected	App	**Confirmed:** Inspector's own paired login on App, capture-and-checklist pattern, same as the rest of the field roles.
S8.6	Confirm all DA conditions addressed	PM	—	Portal	checklist against Stage 7's approved conditions

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S8.7	Confirm Home Warranty Insurance in place	PM	—	Portal	requires document
S8.8	CC issued → hold point cleared, construction can commence	System	—	—	`is_hold_point = true`, blocks Stage 9/10

B — APPROVALS & CONTRACT

STAGE 9 CONSTRUCTION TENDER PREPARATION

DEPOSIT MILESTONE

> Corrected from the source matrix, which had PM tendering directly to trade subcontractors — that describes PM acting as his own head contractor, which contradicts the Developer→PM→Builder→Subcontractor→Tradie chain this whole design is built around. Corrected so PM tenders to **Builders**; it is the Builder who subsequently engages trade subs, in his own scope. No schema change needed — `contracts.party_type='subcontractor'` already covers a Builder as the largest-scope subcontracted entity.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S9.1	Prepare detailed Bill of Quantities from Stage 6's final drawings + Stage 5's tier	PM/Estimator	—	Portal	the real construction-phase Cost Plan begins here, distinct from the DA-phase consultant costs tracked since Stage 1
S9.2	Package tender documents (BOQ, drawings, specs)	System	—	—	
S9.3	Issue tender to a shortlist of Builders	PM	—	Portal	
S9.4	Collect and compare Builder tenders	PM	—	Portal	
S9.5	Select winning Builder — commercial decision, value locked	PM	—	Portal	
S9.6	**Job Award** — formal invitation sent to the selected Builder	PM	—	Portal	sourced from PM's existing contact book (Introduction already happened, possibly long before this project existed — see Section 1.1); this is the formal, later, project-specific step, not the first time PM and Builder have interacted

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S9.7	Builder reads and accepts or declines — lightweight in-app tap from his already-verified account	Builder	—	App/Portal	**Resolved:** no separate signature ceremony — Builder's identity is already a government-backed credential, verified at Self-Registration and displayed via VeriTrade, so the tap doesn't need to carry identity-proving weight itself. What actually makes acceptance hard to repudiate is S9.9's deposit transaction, not this tap alone (see S9.9's note) — the two are a pair, not independent steps. **Remaining legal question, not a design choice:** whether an in-app tap satisfies what some states' home building contract legislation requires for a validly executed contract, or whether a compliant document + proper e-signing flow is still needed alongside it — confirm with a solicitor before this ships.
S9.8	Handover — technical documents (architectural, structural, DA conditions — no pricing) become available to Builder's crew	System	—	—	fires on S9.7 accept; must be pulled into the App's offline sync cache before the crew heads to a signal-dead site, not fetched on demand
S9.9	Finalize and lock the construction Cost Plan	PM	—	Portal	— **BILLING MILESTONE: Deposit**, once contract signed. This deposit invoice is what actually makes S9.7's acceptance binding in practice — a TPAR-reportable payment against Builder's own ABN is a far stronger anchor than a tap alone, and is the evidence a repudiation claim would have to contend with.
S9.10	Seed `project_stages` 10–18 + `tasks` from the stage template, now that a Builder is attached	System	—	—	this is where the Gantt actually comes alive. For a multi-unit project, this is also where the Builder sets task-level `scope` (project | unit) during job breakdown — system proposes sensible defaults, Builder confirms/overrides each one.

C — SITE & STRUCTURE

STAGE 10 SITE WORKS EARTHWORKS

> First stage to drop PM from the actor line entirely, and first stage marked App, not Portal — the real handover from desk to field.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S10.1	Site tracking module activates	System	—	—	fires once CC is issued (Stage 8) and Builder is engaged (Stage 9)
S10.2	Site clearing and demolition (if applicable)	Foreperson	Site Supervisor	App	
S10.3	Cut and fill earthworks to design levels	Foreperson	Site Supervisor	App	
S10.4	Install temporary site amenities (toilet, power, water)	Foreperson	Site Supervisor	App	
S10.5	**HOLD POINT** — Set out building position (surveyor pegs)	Surveyor (external) + Foreperson records	Site Supervisor	App	corrected per external review: elevated from recorded evidence to a hard `is_hold_point` blocking Stage 11. Cost asymmetry drove the decision — verifying setout here is cheap; discovering a wrong boundary/ setback at Stage 12's slab pour is not.
S10.6	Erect temporary fencing	Foreperson	Site Supervisor	App	

C — SITE & STRUCTURE

STAGE 11 FOUNDATION SLAB IN GROUND SERVICES

> First stage where `tradie` appears in the actor line — the full tick-then-verify chain runs for real from here on: a device-writable status field (tick) plus a protected `verified_by` / `verified_at` field settable only by the next role up (verify), same discipline as `is_validated` .

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S11.1	Excavation for footings + underground services	Tradie	Foreperson	App	
S11.2	Underground drainage (sewer, stormwater)	Tradie	Foreperson	App	
S11.3	Water supply rough-in	Tradie	Foreperson	App	
S11.4	Gas lines (if required)	Tradie	Foreperson	App	
S11.5	Electrical conduit and services	Tradie	Foreperson	App	
S11.6	Reinforcement steel (rebar)	Tradie	Foreperson	App	
S11.7	Formwork for concrete slab	Tradie	Foreperson	App	
S11.8	Site Supervisor checks services — correct location and depth	Site Supervisor	—	App	
S11.9	**HOLD POINT** — Electrician's compliance certificate (rough-in)	Inspector (Electrical)	protected	App	**Resolved, per external review:** elevated from a single PC visit to three named items, matching Stage 12's multi-document hold-point pattern
S11.10	**HOLD POINT** — Plumber's compliance certificate (rough-in)	Inspector (Plumbing)	protected	App	
S11.11	**HOLD POINT** — Principal Certifier's overall verification	Inspector	protected	App	PC's sign-off reviews the two trade certificates above, doesn't replace them
S11.12	Concrete pour	Tradie	Foreperson	App	gated on S11.9–S11.11

C — SITE & STRUCTURE

STAGE 12 CONCRETE SLAB POUR FOUNDATION

BASE MILESTONE

> First stage to reveal that a hold point is a *checklist*, not a boolean — three named, partly-conditional documents, not one sign-off. This generalises: `hold_point_requirements` is a list per stage, each item flagged `statutory` or `internal_qa`, each carrying its own `required_inspector_type` where relevant.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S12.1	Pour ground floor slab and footings	Foreperson*	Site Supervisor	App	*actor line names only Site Supervisor + Inspector for this stage — defaulted Foreperson still executes; flag if this is really an external concrete-pour subcontractor with no login, same pattern as earthworks
S12.2	Finish concrete to design levels	Foreperson*	Site Supervisor	App	
S12.3	Install anchor bolts and connection plates	Foreperson*	Site Supervisor	App	
S12.4	Cure concrete	System	—	—	timer, not an action — logs cure-start date
S12.5	Waterproofing (if required)	Foreperson*	Site Supervisor	App	
S12.6	Record concrete delivery dockets	Site Supervisor	—	App	
S12.7	**HOLD POINT** — Form BA2 (slab inspection)	Inspector	protected	App	
S12.8	**HOLD POINT** — Core compression test results	Inspector	protected	App	conditional (“if required”). **Resolved — hybrid:** auto-flagged as required when Stage 2's hazard audit (S2.4) already recorded the triggering soil/hazard condition; falls back to manual Inspector on-the-day judgement only where S2.4 didn't capture a relevant finding.
S12.9	**HOLD POINT** — Compaction test log	Inspector	protected	App	conditional, same hybrid resolution as S12.8
S12.10	Stage advances only once S12.7–S12.9 satisfied	System	—	—	— **BILLING MILESTONE: Base**

C — SITE & STRUCTURE

STAGE 13 FRAME ROOF CONSTRUCTION

FRAME MILESTONE

First stage requiring two *different kinds* of independent sign-off, not the same Inspector role twice — a structural engineer certifying adequacy and a building surveyor certifying compliance check different things and neither substitutes for the other. `hold_point_requirements` needs a `required_inspector_type` per line for exactly this reason. Also the first explicit auto-chaining rule in the matrix, generalised as a standing pattern: hold point cleared → downstream stage scheduled → relevant trades notified.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S13.1	Erect structural frame (timber or steel)	Tradie	Foreperson	App	
S13.2	Install structural bracing	Tradie	Foreperson	App	
S13.3	Install roof trusses and sheathing	Tradie	Foreperson	App	
S13.4	Install roof cladding (tiles or metal)	Tradie	Foreperson	App	
S13.5	Install gutters and downpipes	Tradie	Foreperson	App	
S13.6	Install ceiling insulation	Tradie	Foreperson	App	
S13.7	Record timber/steel + roof material delivery dockets	Site Supervisor	—	App	
S13.8	**HOLD POINT** — structural engineer inspects frame	Inspector (Engineer)	protected	App	
S13.9	**HOLD POINT** — building surveyor inspects frame	Inspector (Surveyor)	protected	App	
S13.10	**HOLD POINT** — F5 timber grade certification on file	System (document check)	—	—	
S13.11	Stage validated → auto-schedules Stage 14 + auto-notifies electrical/plumbing trades	System	—	—	— **BILLING MILESTONE: Frame**

D — FIT-OUT & FINISHES

STAGE 14 LOCK UP EXTERNAL WORKS

LOCK-UP MILESTONE

> Corrected: the source matrix lists “install plasterboard lining” here, which conflicts with Stage 15's hold point requiring electrical/plumbing inspection *before* wall lining — Stage 15 comes after Stage 14, so lining would already be up by the time the inspection it must precede occurs. Relocated to the tail of Stage 15, gated on that stage's hold point. Stage 14 stays scoped to the external envelope only, which is what “lock-up” and “secured from weather” actually mean. Also the first stage with no statutory hold point at all — its inspections are internal QA, not blocking.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S14.1	Install external wall cladding	Tradie	Foreperson	App	
S14.2	Install windows and glazing	Tradie	Foreperson	App	
S14.3	Install external doors	Tradie	Foreperson	App	

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S14.4	Install wall insulation	Tradie	Foreperson	App	
S14.5	Window/cladding QA check — internal, not statutory	Site Supervisor	—	App	`inspection_type='internal_qa'`, doesn't block progress
S14.6	Window energy rating certificate on file	System	—	—	
S14.7	Confirm building secured from weather	Site Supervisor	—	App	— **BILLING MILESTONE: Lock-up**

STAGE 15 INTERNAL SERVICES ROUGH IN

> Plasterboard lining relocated here from Stage 14 (see above) and gated on this stage's hold point — services must be inspected while the wall cavity is still open.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S15.1	Electrical rough-in (cabling, switchboard, smoke alarms)	Tradie	Foreperson	App	
S15.2	Plumbing rough-in (internal drainage, water supply)	Tradie	Foreperson	App	
S15.3	HVAC rough-in (ducting, refrigerant pipes)	Tradie	Foreperson	App	
S15.4	Data/comms cabling (if required)	Tradie	Foreperson	App	
S15.5	**HOLD POINT** — electrical inspection	Inspector (Electrical)	protected	App	
S15.6	**HOLD POINT** — plumbing inspection	Inspector (Plumbing)	protected	App	
S15.7	Electrical safety compliance certificate on file	System	—	—	
S15.8	Install plasterboard / wall lining	Tradie	Foreperson	App	relocated from Stage 14; gated on S15.5 + S15.6

STAGE 16 FIT OUT FINISHES

FIXING MILESTONE

> No statutory hold point — internal QA only, same pattern as Stage 14. First stage where the deeper room-level task nesting (Unit → Room → Trade item, via `tasks.parent_id`) actually earns its keep: tiling, for instance, is bathroom 1, bathroom 2, kitchen splashback, not one flat per-unit task.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S16.1	Plastering and setting (cornices, stopping)	Tradie	Foreperson	App	distinct trade/timing from S15.8's sheet-hanging
S16.2	Painting	Tradie	Foreperson	App	
S16.3	Cabinetry installation	Tradie	Foreperson	App	
S16.4	Tiling	Tradie	Foreperson	App	room-level nesting applies
S16.5	Flooring	Tradie	Foreperson	App	
S16.6	Fixtures and fittings	Tradie	Foreperson	App	
S16.7	**HOLD POINT** — Light fittings, switches, power points; energisation requires an independent electrical safety certificate	Tradie	Electrician (licensed) + Inspector	App	corrected per external review and confirmed by research: every state/territory requires a distinct certificate before mains energisation, separate from Stage 15's rough-in inspection — named differently per jurisdiction (COES in VIC/WA, CCEW in NSW, Certificate of Testing & Compliance in QLD, CoC in NT/SA, CES in ACT/TAS). `hold_point_requirements` should carry a `jurisdiction` field so the correct certificate name/requirement is injected automatically based on the project's state, same conditional-requirement pattern as Stage 2's BAL report and Stage 12's compression/compaction tests.
S16.8	Appliance installation	Tradie	Foreperson	App	

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S16.9	Record deliveries (cabinetry, tiles, fixtures, appliances)	Foreperson	—	App	
S16.10	Final trade sign-off, per room	Foreperson	Site Supervisor	App	— **BILLING MILESTONE: Fixing**

D — FIT-OUT & FINISHES

STAGE 17 EXTERNAL WORKS LANDSCAPING

> Lightest stage in the matrix — no hold point, no named certificates, purely internal QA.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S17.1	External cladding final finishes and painting	Tradie	Foreperson	App	
S17.2	Driveway and paths	Tradie	Foreperson	App	**Resolved, with a caveat:** same conditional `jurisdiction`-field mechanism as S16.7, but unlike S16.7's electrical certificate (verified as a consistent state-level rule), a crossover/ stormwater permit is more likely to vary by *local council*, not just state — potentially hundreds of distinct requirements, not eight. Kept as a PM-checked manual flag per council rather than claimed as a fully system-verified national table.
S17.3	Retaining walls	Tradie	Foreperson	App	
S17.4	Boundary fencing	Tradie	Foreperson	App	
S17.5	Hard landscaping (paving, garden beds)	Tradie	Foreperson	App	
S17.6	Soft landscaping (turf, plants, trees)	Tradie	Foreperson	App	
S17.7	Final clean (builders clean)	Tradie	Foreperson	App	
S17.8	Record deliveries (fencing, retaining wall materials, turf, plants)	Foreperson	—	App	

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S17.9	Site Supervisor final external walk-through	Site Supervisor	—	App	

E — COMPLETION & HANDOVER

STAGE 18 COMPLIANCE HANDOVER ASSET DEPRECIATION

COMPLETION MILESTONE

Where the two biggest cross-cutting threads of the whole design either pay off or reveal a real gap. S18.10 is the payoff of the entire hold-point architecture — the system finally cross-checking that Stage 12 and Stage 15 were genuinely validated back when they happened, which is only meaningful because `is_validated` has been a protected, unforgeable column since Stage 8. S18.16 is the payoff of the Verified Work History thread and the hand-off point into VeriTrade — every participant's project-completion attestation lands here. S18.11 was flagged as a genuine defect (auto-firing ATO lodgement with no human review) and is corrected below by splitting it into a prepare step and a mandatory approval gate, per external review.

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S18.1	Final building inspection by Principal Certifier	Inspector	protected	App	
S18.2	Issue Form BA3 (Certificate of Compliance)	Inspector	protected	App	
S18.3	**HOLD POINT** — Issue Occupation Certificate	Inspector	protected	App	
S18.4	Electrical safety certificate on file	System	—	—	
S18.5	Plumbing compliance certificate on file	System	—	—	
S18.6	Practical Completion walkthrough — defect list generated	PM & Client	—	App (PM) / Portal (Client)	
S18.7	Defect rectification (if any)	Tradie	Foreperson	App	loops until clear
S18.8	Client handover — keys, manuals, warranties	PM	—	App/Portal	
S18.9	Client satisfaction survey	Client	—	Portal	
S18.10	System cross-check: Stage 12 + Stage 15 actually `is_validated`	System	—	—	blocks S18.11 if either failed — the payoff of the whole hold-point architecture

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S18.11	System prepares fixed asset entry + depreciation schedule as a draft — not lodged	System	—	—	figures only; no external transmission at this step
S18.12	**APPROVAL GATE** — accountant/BAS agent reviews and approves before ATO lodgement fires	Accountant/BAS Agent	protected	Portal	corrected per external review: closes the gap flagged in v2.0 — the single most consequential external event in the whole matrix no longer fires without independent, licensed human review. This role is new to the matrix; likely an external, no-login party like the Structural Engineer/ Surveyor, with PM uploading their sign-off, unless volume justifies a dedicated paired login.
S18.13	Auto-email handover pack (Form BA3, OC, manuals, as-built record)	System	—	—	the as-built record is a query over accumulated evidence since Stage 10, not a new authoring task — needs its own long/indefinite retention class, separate from any routine log-prune policy
S18.14	Client final digital sign-off	Client	protected	Portal	frozen/hashed, same discipline as Stage 5 and Stage 7
S18.15	`project.status = 'completed'`	System	—	—	— **BILLING MILESTONE: Completion**

CODE	SUB-STEP	RESPONSIBLE	VERIFIER	SURFACE	NOTES
S18.16	Attestation emitted to every participant's Verified Work History	System	—	—	the payoff of the testimony/ evidence architecture — every tradie, foreperson, Site Supervisor, Builder and Inspector who touched this project gets a project-completion record added to their own portable history. This is the hand-off point into **VeriTrade** (the B2B trade-verification platform, specified separately) — subject in every case to that individual's own opt-in publishing choice (Section 4.2).

Part 3 — Open Questions & Unresolved Gaps

Consolidated from the per-stage flags in Part 2. Each entry below was raised during design and either deliberately deferred, left as a founder decision, or identified as a genuine defect worth fixing before the software specification is finalised. Nothing here should be silently assumed one way or the other when Part 2 is handed to engineering.

Design decisions still open

1. **Stage 9 —** Single Builder per project is the assumed default throughout. A single project genuinely requiring two concurrent, independent Builder engagements (e.g. a shell contractor and a separate fit-out contractor) was raised as a possibility and explicitly not designed — would be a materially bigger architectural change than the roster/panel concept. Kept as the v1 default deliberately, same resolution as the multi-unit Gantt question above — not designed for until a real project needs it.

Resolved following external and founder review

1. **Stage 18, S18.11 — Fixed.** Split into S18.11 (system prepares the fixed asset entry and depreciation schedule as a draft, not lodged) and S18.12 (a mandatory approval gate — an accountant/BAS agent reviews and approves before ATO lodgement fires). The single most consequential external event in the matrix no longer fires without independent, licensed human review.
2. **Stage 10, S10.5 — Resolved.** Survey set-out elevated from recorded evidence to a hard `is_hold_point` blocking Stage 11 — the cost asymmetry (cheap to verify now, potentially project-ending to discover at Stage 12's slab pour) settled the question.
3. **Stage 16, S16.7 — Resolved and confirmed by research.** Every Australian state/territory requires a distinct certificate before mains energisation (named differently per jurisdiction — COES, CCEW, Certificate of Testing & Compliance, CoC, CES). `hold_point_requirements` gains a `jurisdiction` field so the correct certificate is injected automatically per project state, using the same conditional-requirement pattern already built for Stage 2's BAL report and Stage 12's compression/compaction tests.
4. **Section 1.8, Multi-unit Gantt/Line-of-Balance — Resolved.** Not a roadmap-priority call between building single-house or multi-unit tooling first — S1.3 now has the PM declare unit count at project creation, and that value conditionally selects the rendering: standard bar-chart Gantt for one unit, Line-of-Balance flow view for more than one. The LOB component itself still needs building, but the trigger is now exact (the first multi-unit project created) rather than an open question about engineering sequencing.
5. **Stage 3, revise/regenerate loop — Resolved.** Built as S3.10/S3.11 rather than deferred — concept revision is likely to hit at the highest-leverage sales moment (client wants it different), so it wasn't worth leaving as a future problem. Same versioning discipline as every other frozen record: new version via `supersedes_id`, original never overwritten.
6. **Stage 4, response-clock tracking — Resolved.** Added as S4.4, same mechanism as Stage 7's statutory countdown, reused rather than left inconsistent between the two stages.
7. **Stage 7, rejection/revision path — Resolved.** A declined DA sign-off at S7.3 now explicitly routes back to S6.2, billed as a Variation against the architect's redraft cost — reuses the existing Variation mechanism rather than inventing a new one.
8. **Stage 8, Inspector's device-pairing surface — Confirmed, not just defaulted.** App, capture-and-checklist pattern, same as the rest of the field roles.

9. **Section 1.1, Introduction PIN/QR mechanic — Resolved.** Defaults to each party's own ordinary account login, not a fresh out-of-band PIN — lower friction, consistent with how a "connect" action works elsewhere. Revisit only if proximity fraud at this step becomes a real, observed problem, not a theoretical one.
10. **Stage 9, Job Award signature mechanic — Resolved.** A lightweight in-app tap, not a separate signature ceremony — Builder's identity is already a government-backed credential verified at Self-Registration. What actually makes the acceptance binding is explicitly linked to S9.9's deposit transaction (TPAR-reportable against Builder's own ABN), not the tap in isolation. One genuine legal question remains open, not a design choice: whether this satisfies what some states' home building contract legislation requires for contract execution — confirm with a solicitor before this ships.
11. **Stage 11, S11.9 hold point — Resolved.** Split into three named items (S11.9 electrician certificate, S11.10 plumber certificate, S11.11 PC overall verification), matching Stage 12's proven multi-document hold-point pattern rather than the single-visit default kept purely for convenience.
12. **Stages 2, 12 & 13, conditional hold-point triggers — Resolved — hybrid.** Auto-flagged where Stage 2's hazard audit (S2.4) already recorded the triggering condition (soil classification, bushfire zone); falls back to manual Inspector judgement only where Stage 2 didn't capture a relevant finding. Not forced to a single answer where the data already answers it part of the time.
13. **Stage 17, driveway/council sign-off — Resolved, with an honest caveat.** Same conditional `jurisdiction`-field mechanism as S16.7, but flagged as likely varying by local council rather than state — kept as a PM-checked manual flag, not claimed as a fully system-verified national table the way S16.7's state-level certificate could be.

Larger architectural items, deliberately deferred rather than designed

1. **Cross-organisation tenant isolation** for an arms-length Builder relationship (as opposed to the same-tenant model used for the founder's own first project). Real, outlined in principle (an attested data slice crossing a tenant boundary, mirroring the QR trust-exchange mechanism), but not built — needed only once a genuinely independent, externally-owned Builder business is engaged by a different Developer/PM business than the one that built this system.
2. **Public, open, searchable trust directory** (a hipages-style consumer product) vs. verification scoped to users of this platform. Confirmed as scoped-to-platform for now — the public-facing version is a materially different, much larger undertaking (consumer marketing, cold-start network effects) that should wait until the scoped version has accumulated real evidence worth showing.
3. **Material Selections module** (Stage 6) — flagged repeatedly as deserving its own dedicated design pass, comparable to the attention given to Gantt and Cost Plan at the start of this process. Not yet designed in full; only its data shape (pricing model, freeze points, variance band) has been fixed.
4. **Minors on the platform via TAFE/school distribution** — needs a dedicated privacy/consent design (parental consent, data minimisation, working-with-children considerations for a permanent, evidence-backed professional record), not the adult self-registration flow with an age gate attached.

Items requiring legal/compliance confirmation, not a design decision

Flagged throughout as “check with a solicitor / current legislation” rather than answered definitively, since these are jurisdiction- and time-sensitive facts, not architecture choices:

- Cost-plus contract disclosure obligations under the relevant state's home building contract legislation.
- Statutory response timeframes for progress payment claims under security-of-payment legislation.

- Consent requirements for audio/video capture of people (as distinct from site/work photography) under state surveillance/listening-device legislation, given “recording” was named explicitly as an evidence type.
- Exact structural defect liability period length(s) by state and defect category, which sets the minimum retention period for the as-built handover record (Section 1.11).
- Referral-commission disclosure obligations, in the event the commercial structure ever moves away from the current absorbed-DA-cost model back toward a client-facing fee arrangement.

Part 4 — Ethical Safeguards & Governance

Raised late in the design process, deliberately treated with the same seriousness as any hold point rather than as a closing disclaimer. The system designed in Parts 1–2 is, honestly named, workplace surveillance and accountability infrastructure — every tick, geofenced photo, and verified/unverified flag is one party's recorded judgement of another, in a relationship where the two parties rarely hold equal power. That is not automatically wrong — a verified, evidence-backed work history is a genuinely more inclusive credentialing mechanism than paper qualifications alone, rewarding demonstrated competence over legible paperwork, which matters most for exactly the workers who tend to lack the latter despite real skill. But the case for the system does not excuse skipping the case against uncritical trust in it. Four safeguards were identified as necessary, not optional, before this ships.

4.1 Escalation & dispute mechanism

The gap: nothing in Parts 1–2 let the *verified party* contest a verifier's judgement. A Foreperson declining to verify a Tradie's tick is unforgeable and permanent under the frozen/hashed pattern (Section 1.4) — but that same discipline, with no appeal path for the person being judged, just moves the "he said/she said" problem up one level and gives one side a hash to back their version.

Resolution: a Tradie who wishes to contest a verification decision escalates **Site Supervisor first, decisively** — not Builder. Site Supervisor is independent of Builder by the locked role model (Section 1.1: PM's own appointee, not a subset of Builder's role), so this is a genuine outside check, not asking a colleague to referee. Builder is only involved if Site Supervisor is unavailable or himself conflicted — Builder is never the final word on a dispute against his own Foreperson's judgement, which would be asking him to referee his own team.

The dispute itself is an in-app action, not a verbal complaint: the Tradie flags the specific record and attaches his own counter-evidence. Per the append-only discipline used everywhere else in this design, the original non-verification is never erased once a dispute resolves — the record shows both the original call and the resolution, the same correction-via-new-version pattern as every other frozen record in Part 1.

4.2 Portability vs. public visibility

The gap: "his own advertising" (Section 1.7) is only true once a worker can actually take his history with him — and the same-tenant v1 model (Section 1.1) means it currently lives inside one company's database.

Resolution, corrected: this was originally written against an assumed open, consumer-facing directory (hipages-style, any homeowner browsing). The platform since named for this — **VeriTrade** — is explicitly **B2B-gated**, not open-public: commercial builders, tier-1 contractors, and specialised trade businesses vetting talent, not homeowners reading reviews. That is a materially lower-exposure profile than what this section originally assumed, and the wording is corrected accordingly. "Public" and "portable" are still kept as two separate settings, not one — the underlying record stays **private by default and portable** (the worker can show it to whoever he is engaging with), and appearing on VeriTrade is an **explicit, revocable opt-in**, the worker's choice to advertise, not something the platform does to him automatically. The lower exposure of a gated B2B audience versus the open internet reduces, but does not remove, the stakes here — a wrongly-declined verification visible to a pool of prospective commercial employers is still real

reputational risk, which is exactly why Section 4.1's dispute mechanism has to be solid before any default-visible listing on VeriTrade would be safe.

4.3 Minimising the invoice-gate's surveillance footprint

The gap: tying payment to in-system proof of task completion (the adoption mechanism identified early in this design) is consent under economic duress for the worker being tracked, not a freely chosen opt-in — naming it plainly rather than calling it a neutral incentive.

Resolution — harm reduction, not a full fix, and named as such:

- **Minimise what the gate actually needs:** proof of *this specific task* — a geofenced photo and timestamp — nothing more. It must not be allowed to quietly grow into continuous location tracking through the working day; this is a hard technical limit, not a policy promise that can erode.
- **Separate retention from evidence:** not everything captured needs the long retention set for the compliance record (Section 1.11). Granular day-to-day capture can be pruned once its immediate purpose (payment gate, dispute window) has passed — only the validated milestone itself needs to survive for years, shrinking the standing surveillance footprint without weakening the underlying protection.
- **No hidden dossier:** whatever Foreperson and Builder can see about a worker, that worker can see about himself. This does not remove the power imbalance, but it removes the secrecy, which is usually most of what makes tracking feel like surveillance rather than shared record-keeping.
- **Honesty about the limit:** if withholding data means withholding pay, that is not a free choice, and no consent checkbox changes that. Proportionality is a real, partial mitigation — not a solved problem, and should not be marketed as one.

4.4 Account deletion — deactivation, not erasure

The gap: "the user can choose to stop sharing / delete his account" needs to mean different things for different data, or it either breaks a promise or reopens the exact gameability problem the evidence architecture exists to close.

Resolution: profile details, contact info, and anything self-uploaded and discretionary are deletable outright. The record of a *specific, already-relied-upon completed job* is not solely the worker's data to erase — a Foreperson verified it, it may have fed a hold point, it may matter to a future warranty or insolvency dispute, and other real parties have a legitimate, ongoing stake in it continuing to exist. Allowing outright deletion of a bad outcome before a future job application would reopen, via erasure, the same integrity problem this whole system was built to close via evidence.

"Delete account" is therefore **deactivation**: all future capture stops immediately; the profile is pulled off any public directory at once (reversing Section 4.2's opt-in is always immediate and unconditional); anything purely personal is removed. Historical, already-relied-upon project evidence is retained under the same policy as everything else, no longer tied to an active, growing profile. This must be disclosed to the user plainly and explicitly at the point of deletion — what is removed, what is retained, and why — not as a quiet asterisk in a settings screen, because the alternative is promising something the system cannot actually

deliver without breaking the protection it exists to provide everyone, including the person deleting their account.

4.5 Named, not yet resolved

Two further asymmetries were raised and are recorded here as things worth revisiting, not yet designed: the commercial-visibility walls (Section 1.9) protect Builder's and PM's margins from the Client, who is typically the least sophisticated, least repeat-player party in the chain, making the largest purchase of his life — the system's protective architecture currently runs more in the builder/PM direction than the client's. And the AI confirm-gate pattern (Section 1.2 origin) is only a meaningful safeguard for as long as the confirming human retains the expertise to actually catch an error in what they are confirming — if years of rubber-stamping erode that judgement, the safeguard becomes procedural rather than real. Neither is fatal; both are cheap to design for now and expensive to discover later.

Part 5 — Ready for the Next Phase

This document is the foundation, not the specification itself. Parts 1 and 2 together define, for every one of the 18 stages and their sub-steps: who acts, who verifies, which surface they use, what evidence is produced, which points are hard compliance gates versus internal quality checks, and which stages carry a client billing event. Part 3 is the honest ledger of what still needs a decision, a fix, or outside confirmation before implementation begins in earnest.

The natural next step is translating this into the concrete specification documents this was always aiming at:

Deliverable	Draws primarily from
App specification	Sections 1.2, 1.3, 1.7 — every App-tagged sub-step in Part 2
Portal specification	Sections 1.2, 1.8, 1.10 — every Portal-tagged sub-step, job breakdown/scheduler/ monitoring loop
Dashboard specification	Section 1.7 job-monitoring loop, Section 1.9 visibility walls per role
Server specification	Sections 1.1, 1.4, 1.5, 1.6 — role enforcement, protected columns, hold-point engine, event-chaining
RDBMS schema	All of Part 1's named tables/fields, plus every stage's Data Created / document requirements in Part 2
VeriTrade specification (new, separate deliverable)	Section 1.7 (self-registration, licence, work evidence), Section 4.2 (portability/opt-in), S18.16's hand-off — a B2B-gated professional network built on evidence this system produces, not a stage within it

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Compiled from the full stage-by-stage design conversation between the founder (software engineer, property developer, and PhD candidate researching Vietnam's informal economy) and Claude, covering the complete 18-stage residential construction lifecycle, its sub-steps, and the cross-cutting platform architecture built up around them. Document generated 26 July 2026.